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This document lists the implemented use cases of the Air Ticket Reservation System and the SQL queries executed by each component.

A. PUBLIC / NOT LOGGED-IN USE CASES

1. Public Home Page

Description:

When the user is not logged in, the home page displays links for public search, customer login, airline staff login, customer registration, and airline staff registration.

Main file:

- index.php

No SQL query is directly executed on this page.

2. Search Future Flights

Description:

Allows any user, whether logged in or not, to search for future flights by departure airport, arrival airport, and date.

Main file:

- customer_search.php

Query used:

```
SELECT *  
FROM Flight  
WHERE departure_airport = ?  
AND arrival_airport = ?  
AND DATE(departure_datetime) = ?  
AND departure_datetime > NOW();
```

This query searches the Flight table for future flights matching the selected departure airport, arrival airport, and travel date.

3. Customer Registration

Description:

Allows a new customer to register for an account.

Main file:

- register_customer.php

Query used:

```
INSERT INTO Customer  
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?);
```

This query inserts a new customer record into the Customer table with all required customer attributes.

4. Airline Staff Registration

Description:

Allows a new airline staff member to register for an account.

Main files:

- register_staff.php

Queries used:

```
INSERT INTO Airline_Staff  
VALUES (?, ?, ?, ?, ?, ?, ?);
```

```
INSERT INTO Staff_Phone  
VALUES (?, ?);
```

The first query inserts the new airline staff member into the Airline_Staff table.

The second query inserts a phone number for that staff member into the Staff_Phone table.

5. Customer Login

Description:

Allows a registered customer to log in using email and password.

Main files:

- login_customer.php

- login_auth_customer.php

Query used:

```
SELECT *  
FROM Customer  
WHERE email = ?  
AND password = ?;
```

This query checks whether a customer exists with the given email and password. If a match is found, login succeeds and a session is started.

6. Airline Staff Login

Description:

Allows an airline staff user to log in using username and password.

Main files:

- login_staff.php
- login_auth_staff.php

Query used:

```
SELECT *  
FROM Airline_Staff  
WHERE username = ?  
AND password = ?;
```

This query checks whether an airline staff record exists with the given username and password. If a match is found, login succeeds and session variables are created.

B. CUSTOMER USE CASES

1. Customer Home Page

Description:

Displays the home page for a logged-in customer and shows links to customer-specific features.

Main file:

- customer_home.php

No SQL query is directly executed on this page.

2. View Future Flights

Description:

Shows future flights purchased by the currently logged-in customer.

Main file:

- customer_future_flights.php

Query used:

```
SELECT f.*
```

```
FROM Ticket t
JOIN Flight f
  ON t.airline_name = f.airline_name
 AND t.flight_number = f.flight_number
 AND t.departure_datetime = f.departure_datetime
WHERE t.customer_email = ?
  AND f.departure_datetime > NOW()
ORDER BY f.departure_datetime;
```

This query joins Ticket and Flight to show all future flights associated with the currently logged-in customer.

3. View Past Flights

Description:

Shows past flights purchased by the currently logged-in customer.

Main file:

- customer_past_flights.php

Query used:

```
SELECT f.*
FROM Ticket t
JOIN Flight f
  ON t.airline_name = f.airline_name
 AND t.flight_number = f.flight_number
 AND t.departure_datetime = f.departure_datetime
WHERE t.customer_email = ?
  AND f.departure_datetime < NOW()
ORDER BY f.departure_datetime DESC;
```

This query joins Ticket and Flight to show all past flights associated with the currently logged-in customer.

4. Purchase Ticket

Description:

Allows a logged-in customer to purchase a ticket for a flight if seats are still available.

Main file:

- customer_purchase.php

Queries used:

a. Check seat availability:

```
SELECT a.num_seats, COUNT(t.ticket_id) AS sold
FROM Flight f
JOIN Airplane a
  ON f.airline_name = a.airline_name
  AND f.airplane_id = a.airplane_id
LEFT JOIN Ticket t
  ON f.airline_name = t.airline_name
  AND f.flight_number = t.flight_number
  AND f.departure_datetime = t.departure_datetime
WHERE f.airline_name = ?
  AND f.flight_number = ?
  AND f.departure_datetime = ?
GROUP BY a.num_seats;
```

b. Generate next ticket ID:

```
SELECT COALESCE(MAX(ticket_id), 0) + 1 AS next_id
FROM Ticket;
```

c. Insert purchased ticket:

```
INSERT INTO Ticket
VALUES (?, ?, ?, ?, ?, CURDATE(), CURTIME(), ?, ?, ?, ?);
```

The first query checks whether the number of sold tickets is still less than airplane seating capacity.

The second query generates the next available ticket ID.

The third query inserts the new ticket purchase into the Ticket table.

5. Rate / Comment on Past Flight

Description:

Allows a customer to rate and comment on a flight.

Main file:

- customer_rate.php

Query used:

```
INSERT INTO Rates
VALUES (?, ?, ?, ?, ?, ?);
```

This query inserts the customer's rating and comment into the Rates table for the selected flight.

C. AIRLINE STAFF USE CASES

1. Airline Staff Home Page

Description:

Displays the home page for a logged-in airline staff member and shows links to staff-specific features.

Main file:

- staff_home.php

No SQL query is directly executed on this page.

2. View Flights

Description:

Shows all flights belonging to the airline of the currently logged-in airline staff member.

Main file:

- staff_view_flights.php

Query used:

```
SELECT *  
FROM Flight  
WHERE airline_name = ?  
ORDER BY departure_datetime DESC;
```

This query retrieves all flights operated by the airline associated with the logged-in staff member.

3. Add Airplane

Description:

Allows airline staff to add a new airplane for their airline.

Main file:

- staff_add_airplane.php

Query used:

```
INSERT INTO Airplane  
VALUES (?, ?, ?, ?, ?);
```

This query inserts a new airplane into the Airplane table for the current airline.

4. Create Flight

Description:

Allows airline staff to create a new flight for their airline.

Main file:

- staff_create_flight.php

Query used:

```
INSERT INTO Flight  
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?);
```

This query inserts a new flight into the Flight table with its flight number, times, price, status, airports, and airplane.

5. Change Flight Status

Description:

Allows airline staff to update the status of a flight.

Main file:

- staff_change_status.php

Query used:

```
UPDATE Flight  
SET status = ?  
WHERE airline_name = ?  
    AND flight_number = ?  
    AND departure_datetime = ?;
```

This query updates the status of the selected flight, such as changing it from on-time to delayed.

6. View Customers on a Flight

Description:

Allows airline staff to view all customers booked on a specific flight.

Main file:

- staff_view_customers.php

Query used:

```
SELECT c.name, c.email  
FROM Ticket t  
JOIN Customer c  
    ON t.customer_email = c.email  
WHERE t.airline_name = ?  
    AND t.flight_number = ?
```

```
AND t.departure_datetime = ?  
ORDER BY c.name;
```

This query joins Ticket and Customer to show all customers booked on the specified flight.

D. SESSION / ACCESS CONTROL

1. Logout

Description:

Logs the user out of the application.

Main file:

- logout.php

No SQL query is directly executed.

This component destroys the session and redirects the user to the public home page.

2. Access Control

Description:

Customer-only pages require a valid customer session.

Staff-only pages require a valid airline staff session.

Session variables are checked at the start of protected pages to prevent unauthorized access.